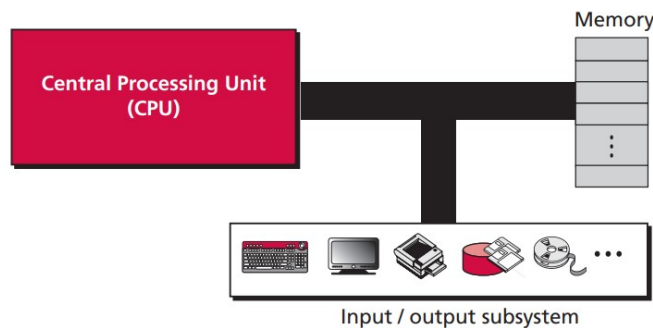


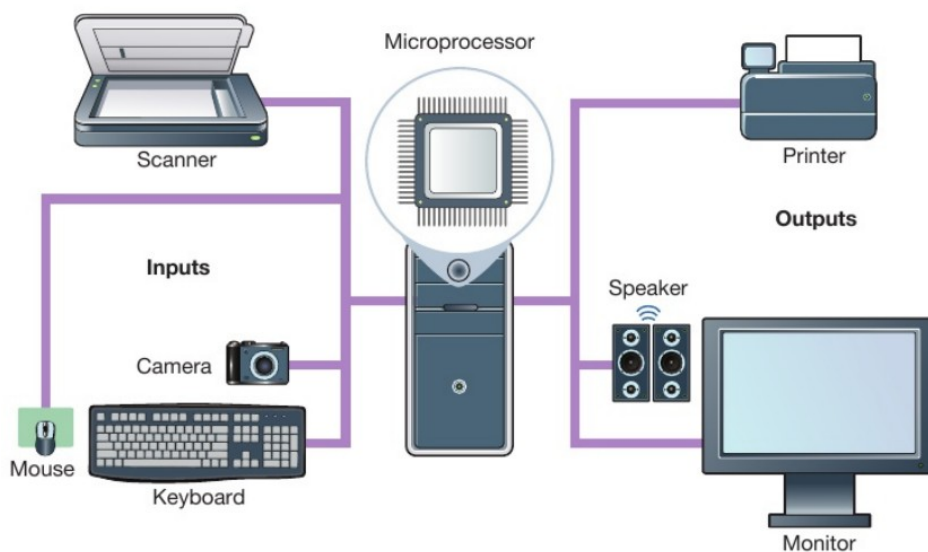
Components of a Computer

The components of a computer can be broadly categorized into three main subsystems: the **Central Processing Unit (CPU)**, the **main memory**, and the **input/output (I/O) subsystem**. These subsystems work together and are interconnected to form a functional, standalone computer. The following sections provide an overview of each subsystem and its role.



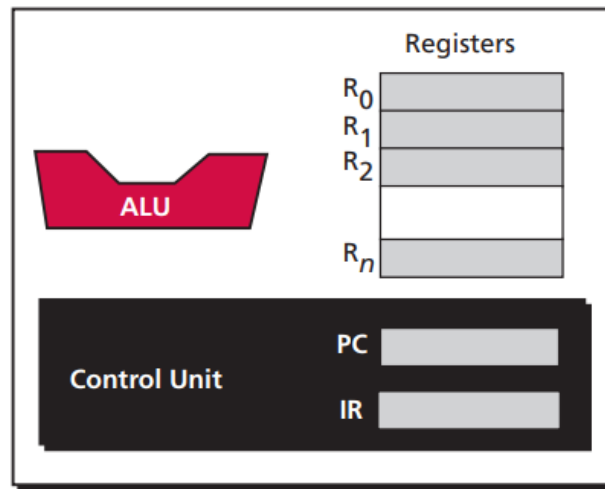
Central Processing Unit (CPU)

The Central Processing Unit (CPU), often referred to as the processor, is the primary component of a computer that executes instructions and performs calculations. In simple terms, the CPU is the hardware that carries out the processing tasks within a computer.



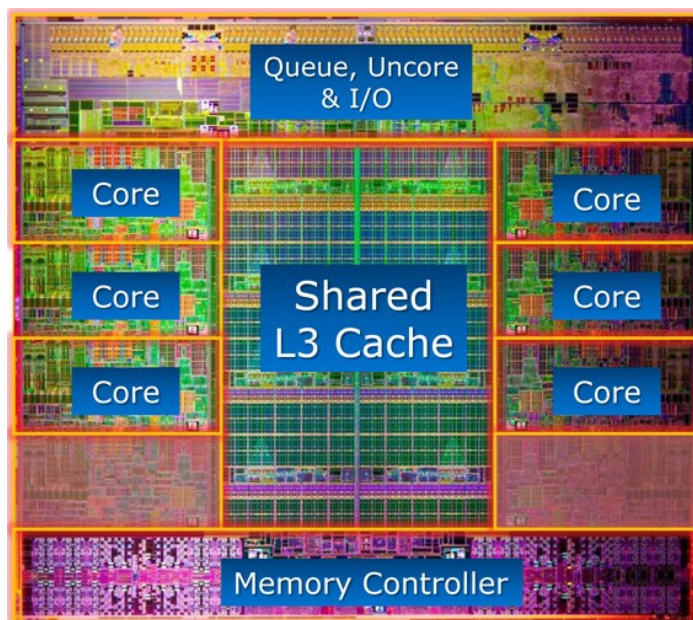
The CPU operates on data and typically consists of three main components:

1. **Arithmetic Logic Unit (ALU):** Handles arithmetic, logic, and shift operations.
2. **Control Unit (CU):** Manages and coordinates the activities of the CPU and other subsystems.
3. **Registers:** High-speed storage locations within the CPU.



R: Register
PC: Program Counter
IR: Instruction Register

Central Processing Unit (CPU)



Intel-core-i7-3960x Die Detail

01. Arithmetic Logic Unit (ALU)

The **Arithmetic Logic Unit (ALU)** is responsible for performing operations on data. These operations include:

- **Logic Operations:** Operations such as **NOT**, **AND**, **OR**, and **XOR** treat input data as bit patterns and produce bit-pattern results.
- **Shift Operations:** Bit patterns can be shifted left or right using logical or arithmetic shifts. Logical shifts are used for unsigned integers, while arithmetic shifts preserve the sign bit, often used for signed integers. These shifts enable quick multiplication or division by powers of two.
- **Arithmetic Operations:** Basic mathematical operations such as addition, subtraction, multiplication, and division.

02. Control Unit (CU)

The **Control Unit (CU)** is the CPU's orchestration system. It generates control signals to manage and synchronize the operations of the ALU, registers, and memory. The CU ensures that instructions are executed in the correct sequence, coordinates data flow between subsystems, and regulates interactions with the I/O subsystem.

03. Registers

Registers are small, fast storage units within the CPU that temporarily hold data during processing. They play a critical role in the efficient operation of the CPU by providing quick access to data and instructions. Registers can be broadly categorized into **general-purpose registers** and **special-purpose registers**, each serving distinct roles:

- **General-Purpose Registers (GPRs):**

Often referred to as **data registers**, these registers are versatile and not limited to specific tasks. They temporarily hold data, intermediate results, and operands during instruction execution. Modern CPUs have multiple general-purpose registers, labeled R1 to Rn, to facilitate faster computations by reducing reliance on slower main memory. GPRs are fundamental to performing arithmetic, logic, and data movement operations.

- **Special-Purpose Registers:**

These registers are designed for specific control and operational tasks, including:

- **Instruction Register (IR):** Temporarily holds the current instruction fetched from memory. The CPU decodes and executes the instruction stored in the IR, allowing the program to progress.
- **Accumulator Register:**
The accumulator is a specialized register used to store the results of arithmetic and logic operations. Many operations, especially in simpler CPU designs, use the accumulator as a central location for processing data. For instance, the result of an addition or multiplication is often stored in the accumulator before being moved elsewhere.
- **Program Counter (PC):** Keeps track of the memory address of the instruction currently being executed. After execution, the PC increments to point to the next instruction in memory.

By combining **general-purpose registers** for flexible data handling and **special-purpose registers** for control and coordination, the CPU achieves high-speed processing and effective program execution.

Processor Core

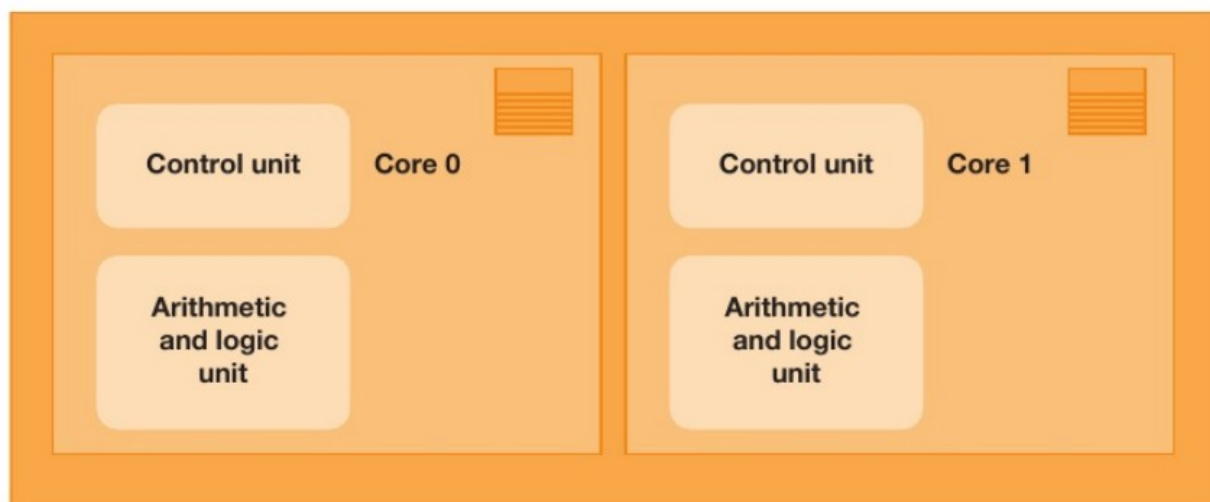
A **core** in a CPU (Central Processing Unit) is an independent processing unit embedded within the CPU chip. Each core is a complete execution engine capable of carrying out instructions and performing calculations on its own. A typical CPU core includes components like:

- **Arithmetic Logic Units (ALUs):** Perform calculations and logic operations.
- **Control Units (CUs):** Direct the operation of the CPU.
- **Registers:** Temporarily store data for quick access.
- **Cache memory:** Stores frequently used data and instructions.

Each core can handle its own set of tasks and execute instructions independently of other cores. This enables the CPU to perform multiple operations simultaneously, improving overall performance and efficiency, especially in **multi-threaded applications**.

Manufacturers introduced **multi-core processors** in 2006 to increase processing speed and improve overall performance. A **multi-core processor** contains more than one processing core, but only a single CPU package. Each core in a multi-core processor is capable of executing instructions independently, which allows the processor to handle multiple tasks simultaneously.

The following diagram illustrates the structure of a **dual-core processor**, which features two cores within a single CPU.



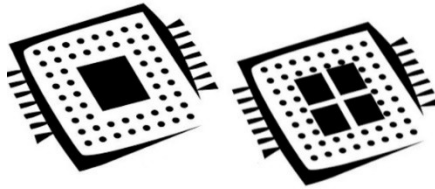
The following table shows the names given to processors having different number of cores.

NUMBER OF CORES	COMMON NAME
1	Single-core
2	Dual-core
4	Quad-core
5	Penta-core
8	Octa(o)-core
10	Deca-core

The advantages of multi-core processors over single-core processors are that the cores can work:

- together of the same program - This is called parallel processing
- on different programs at the same time – This is called multitasking

However , not all programs will run at twice the speed with dual-core processor. The tasks required might not be able to be carried out in parallel. They might be sequential, so that one task requires output from a previous task. This means the second task cannot start until the first has finished.

Single Core vs. Multi core Processors

Originally, processors had a single processing core.

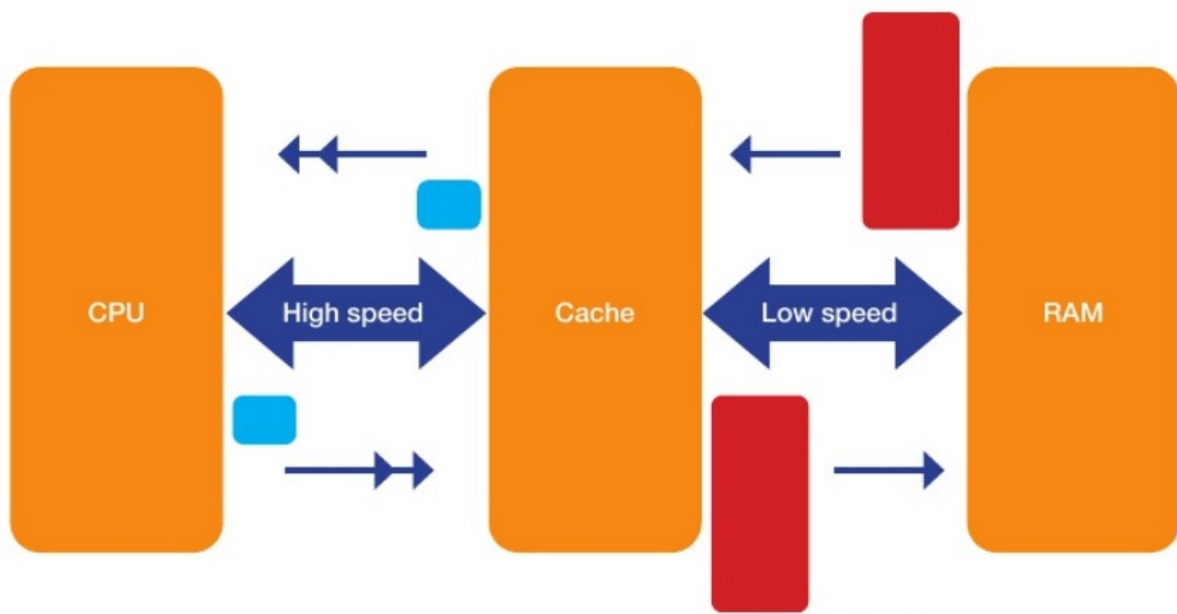
Parameter	Single-Core Processor	Multi-Core Processor
Number of cores on a die	Single	Multiple
Instruction Execution	Can execute Single instruction at a time	Can execute multiple instructions by using multiple cores
Gain	Speed up every program or software being executed	Speed up the programs which are designed for multi-core processors
Performance	Dependent on the clock frequency of the core	Dependent on the frequency, number of cores and program to be executed
Examples	Processor launched before 2005 like 80386,486, AMD 29000, AMD K6, Pentium I,II,III etc.	Processor launched after 2005 like Core-2-Duo,Athlon 64 X2, I3,I5 and I7 etc.

04. Cache Memory

Cache memory is a small, high-speed storage area located inside or very close to the **CPU**. It is used to store frequently accessed data and instructions that the CPU needs quickly. Cache memory is faster than the main system memory (RAM), allowing the CPU to retrieve data more rapidly, which improves overall processing speed.

How Cache Memory Works

When the CPU executes instructions, it often needs data stored in memory. Accessing data from the main memory can be time-consuming, so the CPU uses cache memory to store the most frequently used data or instructions. The idea is that if the CPU needs to access the same data repeatedly, it can retrieve it faster from the cache than from the main memory.



A **cache miss** occurs when the CPU attempts to access data that is not present in the cache. In this case, the processor must fetch the data from a slower memory source, such as the main RAM, which results in a delay in processing.

Cache misses can happen in three main types:

1. **Compulsory Misses:**

These occur when data is being accessed for the first time. Since the data has never been loaded into the cache before, it must be retrieved from the main memory.

2. **Capacity Misses:**

These occur when the cache cannot hold all the data needed by the CPU due to its limited size. As a result, some data may be evicted from the cache, and when it is needed again, the CPU must fetch it from slower memory.

3. **Conflict Misses:**

These occur when multiple pieces of data are mapped to the same location in the cache. This conflict causes some data to be evicted prematurely, leading to a cache miss when it is needed again.

Cache misses can significantly affect CPU performance, as they result in slower data retrieval. Modern CPUs employ advanced algorithms and techniques, like **cache prefetching** and **replacement policies**, to minimize cache misses and enhance performance.

Levels of Cache Memory

Cache memory is typically organized into multiple levels (L1, L2, and L3), with each level having different sizes, speeds, and locations:

1. L1 Cache (Level 1):

- This is the smallest and fastest cache, located directly on the CPU core.
- It stores the most frequently used instructions and data for quick access.
- Typically, L1 cache is divided into two parts: one for instructions (I-cache) and another for data (D-cache).

2. L2 Cache (Level 2):

- L2 cache is larger than L1 but slower.
- It may be located either on the CPU chip or close to it (on the same die but outside the core).
- It stores data that is less frequently used than what is in L1 cache, but still needs to be accessed quickly.

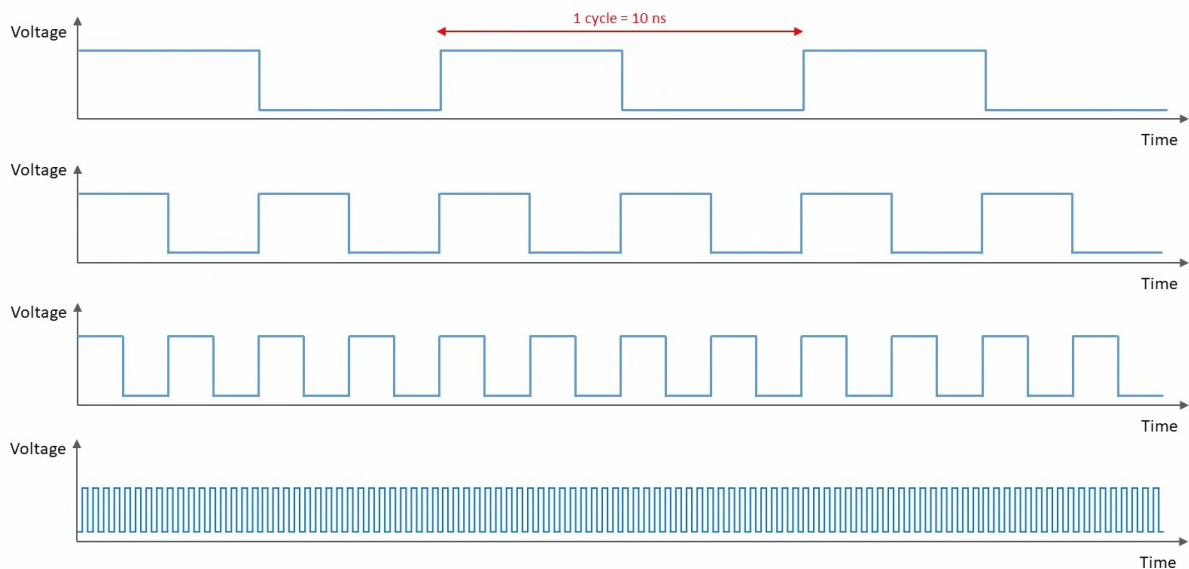
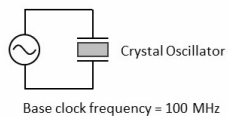
3. L3 Cache (Level 3):

- L3 cache is typically the largest and slowest of the three.
- It is shared by all cores in a multi-core CPU, allowing communication and data sharing between them.
- L3 cache holds data that may not fit in L1 or L2 cache but is still beneficial to store for faster access than going to the main memory.

Cache memory is essential for improving the performance of the CPU by reducing the time it takes to access frequently used data. Without cache memory, the CPU would have to rely on slower main memory, which would drastically decrease processing speeds. By reducing the need to access RAM constantly, cache memory significantly boosts the efficiency of modern processors.

Clock

The **clock** refers to an electronic device that generates a continuous, steady pulse (often called a **clock signal**) used to synchronize the actions of various components, including the **Central Processing Unit (CPU)**.



This clock pulse acts as a timing mechanism, ensuring that all the parts of the computer work in harmony and that operations occur in a well-defined sequence.

The clock generates periodic pulses at a fixed frequency, typically measured in **hertz (Hz)**, where one hertz represents one pulse per second. Modern computers often have clock speeds in the gigahertz (GHz) range, which means billions of pulses per second.

How is the Clock Related to CPU Speed?

The **CPU speed** is closely tied to the clock signal. The clock dictates the rate at which the CPU can execute instructions. Each clock cycle corresponds to a single unit of time during which the CPU can perform specific operations, such as fetching an instruction, decoding it, executing it, or writing results.

Here's how the clock influences CPU speed:

1. Clock Cycles:

Every operation the CPU performs takes one or more clock cycles. For example, a basic instruction like fetching data from memory might take one cycle, while more complex operations like multiplication may take several cycles.

2. Clock Frequency:

The **clock frequency** (or clock speed) determines how many clock cycles occur in one second. A higher clock frequency means more cycles per second, allowing the CPU to process more instructions in a given time period. For instance, a 3.0 GHz CPU can theoretically perform 3 billion cycles per second.

3. CPU Performance:

The CPU's performance is often measured by its clock speed, but the number of clock cycles required to complete an instruction also matters. For example, a CPU with a higher clock speed may not always outperform a CPU with a lower clock speed if the latter executes instructions more efficiently, requiring fewer cycles to complete the same tasks.

4. Overclocking:

Overclocking refers to the practice of increasing the clock speed of the CPU beyond its default specification to achieve higher performance. However, overclocking can lead to increased heat generation and may cause instability if not managed carefully.

Modern day CPU

In earlier CPU designs, components like memory controllers and I/O controllers were separate from the CPU and part of an external chipset. However, modern CPU designs often integrate these components directly onto the chip, a design approach known as System on Chip (SoC). This integration enhances efficiency, simplifies system architecture, and improves overall performance.

Furthermore, many modern CPUs include integrated Graphics Processing Units (GPUs) and other specialized components, which address specific performance, power, and manufacturing needs. As CPU design continues to evolve, the trend toward increased integration is driving the development of more powerful and efficient computing systems.

